



Latest DC-DC Converters For Automotive And Industrial Applications

Some latest DC-DC converters are introduced here, along with the basic and new features, so that designers can select the best devices for their applications

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DC-DC converters are important parts of modern automobiles and industrial applications. This is because direct current (DC) power supplies are used in almost all the electrical devices in automobiles and other industrial applications.

A DC-to-DC converter is basically a DC electric power converter required in some devices because the standard DC sources (mostly batteries like 1.5V, 6V, 12V) available are either too high or too low in voltage to drive a particular circuit or gadget. Therefore some form of power conversion, either step-up or step-down, is required from a standard DC source.

For example, most mobile devices have 3.7V batteries. To charge them from a 12V car battery would require a DC-DC converter circuit because 12V is too high to charge a 3.7V battery. Again, most of the embedded system circuits in automobiles and industrial applications require either 3.3V or 5V DC sources, so you need power converters to drive them from a 12V battery. DC-DC converters are also used in many consumer electronics and domestic electrical systems.



Fig. 1: R05CT05S DC-DC converter (Credit: <https://recom-power.com>)

Power levels of DC-DC converters vary from very low (in small batteries) to very high (for high-voltage power transmission). Most of these converters are available as integrated circuits (ICs) and do not need many external components. These are available as complete hybrid circuit modules or as plug-in modules ready to use in an electronic assembly. Most power converter ICs include an integrated power MOSFET and are classified as regulators because they regulate the output voltage.

This article covers different types of DC-DC converters newly released/announced by different manufacturers for automotive and industrial applications. To better understand the features of these converters, let us first have a look at some major types of DC-DC converters available in the market.

Types of DC-DC converters

The major DC-DC converter types are linear, switched mode, non-isolated, and isolated.

Linear. It uses resistive voltage drop to create and regulate a given output voltage. A linear converter can generate a voltage that is lower than the input voltage.

Switched mode. It converts DC voltage level by storing the input energy temporarily and then releasing that energy to the output at a different voltage. Energy storage can be done in either a magnetic field component like an inductor or a transformer, or in an electric field component such as a capaci-